Everyone knows the fact that (including a toddler):

Thm. Let $f, g : \mathbb{R} \to \mathbb{R}$ be functions which are differentiable at $x = a$. Then, $f+g$ and fg are differentiable at $x=a$ with

$$(f+g)'(a) = f'(a) + g'(a) \quad \text{and} \quad (fg)'(a) = f(a)\cdot g'(a) + f'(a)\cdot g(a).$$

Proof. The first formular is given by:

$$\lim_{h\to 0} \frac{(f+g)(a+h) - (f+g)(a)}{h} \overset{\text{def}}{=} \lim_{h\to 0} \left(\frac{f(a+h)-f(a)}{h} + \frac{g(a+h)-g(a)}{h} \right)$$

$$= \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} + \lim_{h\to 0} \frac{g(a+h)-g(a)}{h}$$

And,

$$\lim_{h\to 0} \frac{(fg)(a+h) - (fg)(a)}{h} \overset{\text{def}}{=} \lim_{h\to 0} \frac{f(a+h)\cdot g(a+h) - f(a)\cdot g(a)}{h}$$

$$= \lim_{h\to 0} \frac{f(a+h)\cdot g(a+h) - f(a+h)\cdot g(a) + f(a+h)\cdot g(a) - f(a)\cdot g(a)}{h}$$

$$= \lim_{h\to 0} \left(f(a+h)\cdot \frac{g(a+h)-g(a)}{h} + g(a)\cdot \frac{f(a+h)-f(a)}{h} \right)$$

$$= \left(\lim_{h\to 0} f(a+h) \right) \cdot \left(\lim_{h\to 0} \frac{g(a+h)-g(a)}{h} \right) + g(a)\cdot \left(\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} \right)$$

$$= f(a)\cdot g'(a) + g(a)\cdot f'(a).$$

Now, we observe this phenomenon from view of algebra.